Chapter 1

Problem, Context and Requirements

1.1 The problem

Floods are Tunisia's deadliest natural hazard: between 1957 and 2018 they caused about 70% of disaster casualties and 60% of economic losses [1], and they cost the country about US$40 million a year [2]. Nabeul, in Cap Bon, is a clear example. On 22 September 2018 about 200 mm of rain fell on Nabeul and up to 225 mm at Beni Khaled, the most since INM records began in 1995; six people died and more than 6,000 families were affected [3], with recovery needs of about US$100 million [1]. In January 2026 some regions had their heaviest rain since 1950, AFP reported five deaths and 350 people rescued, and Nabeul was again among the hardest-hit regions [4].

The warning existed; acting on it failed. A warning was issued on 21 September 2018, the day before the flood [3]. People still died, and some residents stayed on rooftops rather than cross flooded areas to reach evacuation points [3]. The weak step was not the forecast but the last mile: getting a clear, local instruction to the right people and making sure someone acted on it. The World Bank ranks Tunisia 13th of 19 MENA countries for early-warning services and names *inefficient procedures for alert dissemination* among the gaps [1].

Why flash floods need a local approach. In small, steep basins water can rise in less than an hour [5]. Storms are local and gauges sparse, so danger is often seen only when the water is already on the road, and road crossings are where people meet it: in the United States nearly half of flash-flood deaths are vehicle-related [6] (we found no Tunisian figure). The channels can also fail: bulk SMS can be delayed by hours in congested networks, and cell broadcast has no back channel, so nobody knows whether a message was received [7].

Why now. Tunisia's disaster-risk strategy to 2030 calls for people-centred early warning using *simple, low-cost equipment* (Action 8a, priority 1) [8]. National funding goes mainly to INM (US$24 million) [9], and the first pilot sites are Greater Tunis, Ghardimaou and Kebili, not Cap Bon [10]. The UN *Early Warnings for All* initiative aims to cover everyone by the end of 2027 [11].

1.2 Study site: Oued Souhil at the RR27 crossing

We use Oued Souhil, a wadi of about 7.4 km near Nabeul, as the illustrative site (Figure 1.1) [12]. It lies in the governorate hardest hit in 2018 and among the hardest hit in 2026 [3, 4]. It crosses the RR27 primary road on a bridge about 6.15 km along the channel, a concrete place where people meet the water [12]. Satellite imagery shows a sandy bed with industrial buildings beside the channel near the bridge [13], and an agricultural experimental station on the wadi is a possible partner for site access and data [14].

Three nodes are placed along the channel, at about 3.3, 4.9 and 6.15 km. Final node positions, and the choice of crossing itself, will be agreed with the municipality and Civil Protection through a field

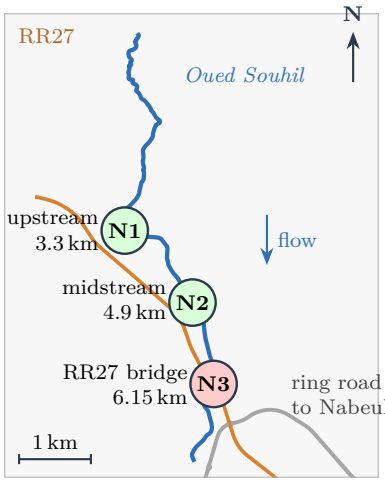


Figure 1.1: Study site: Oued Souhil and the RR27 crossing (plan view, north up), drawn from OpenStreetMap data [12]. Node positions are illustrative and given as distance along the mapped channel: N1 upstream sees the flood first, N2 confirms that the water is moving, and N3 at the RR27 bridge warns people where they would cross.

survey.

1.3 Existing solutions and their gaps

Detecting a flood is no longer the hard part: low-cost LoRa level nodes, SMS alerts and multi-level risk states are all published [15, 16, 17, 18]. What remains open is what happens *after* detection, in places with weak connectivity (Table 1.1).

Table 1.1: Existing solutions: what they do well and what they leave open for a flash flood at a Cap Bon road crossing.

Solution	What it does well	What it leaves open
National SMS alerts (GSMA "We Care") [19]	Reaches citizens through all three operators; alerts decided by the disaster committees.	One-way: no confirmation that anyone received or acted on the alert. No local sensing at a crossing.
National early-warning programme (INM) [1, 9, 10]	Upgrades forecasting and national observation (US$24 M).	National and forecast level; Cap Bon not among first pilots; alert dissemination named as a weakness [1].
Google Flood Hub [20]	Free AI forecasts; Tunisia is covered.	Flash-flood coverage limited to areas with recorded flood history; forecasts risk, does not deliver or confirm a warning.
Low-cost LoRa flood monitors [15, 16, 17, 18]	Level sensing, rate of rise, risk levels, SMS; one solar design works offline with a siren [18].	Alerts stop at "sent": no per-person confirmation, no escalation to municipality or Civil Protection, no link to the national chain. Mostly lab or simulation.
Commercial mass notification (e.g. Everbridge) [21]	Recipients confirm receipt; unanswered alerts escalate.	Subscription-based and internet-dependent, built for institutions, not for a wadi with no network.

The gap. In a search of literature and products between 22 September and 7 October 2026 we found no low-cost flash-flood system that combines a local alarm at the crossing needing no network; confirmation of receipt, even without mobile network; automatic escalation when nobody answers; agreement between independent signals before alarming people; and a bridge into the human-decided national alert channel. Chapter 2 details the comparison (Table 2.1).

1.4 Community need

People at a wadi crossing need a clear, local warning early enough to act, and the responsible authorities need to know that someone has acted on it. Detecting rising water, or sending a message,

is not enough: the system must check that the message reached a person, say what to do, and move the alert up the chain when nobody answers. This serves drivers and pedestrians at the crossing ("do not cross"), residents and workers near the wadi, volunteers and the municipal duty officer, and Civil Protection and the regional committee, who need trustworthy evidence to decide on a public alert.

1.5 Measurable impact

Phase 2 targets are measured on the prototype and match the verification plan of Chapter 2 (Table 2.5); pilot targets apply to a field deployment after the competition (Table 1.2). They measure the warning chain, not lives saved, which a short pilot cannot show.

Table 1.2: Impact metrics and targets.

Metric	What it shows	Target
Phase 2 prototype (bench and replayed floods)		
Detection to beacon	The crossing is warned at once, with no network	$\leq 5\,\mathrm{s}$ after the Danger level is reached
Radio delivery	Readings reach the gateway	$\geq 99\%$ with retries
Escalation on silence	An unanswered alert moves up the chain	Municipality reached within $3\,\mathrm{min} \pm 5\,\mathrm{s}$
Time to acknowledge	Alerts reach a person who responds	Median $< 1\,\mathrm{min}$ in drills
False alarms	One faulty signal cannot alarm people	No Warning from a single signal group
Silence is a fault	A dead sensor is never read as "safe"	Fault reported within 3 heartbeats
Offline operation	The chain survives internet and mobile outage	Beacon fires; mesh acknowledgement accepted
Local repair	Technicians can maintain it	Part swapped using the manual alone
Pilot after the competition (field)		
Reach	Households and sites in the agreed zone	$\geq 90\%$ reached directly or by a volunteer
Response in real drills	The chain works with real people	Median time to acknowledge $< 1\,\mathrm{min}$
Trust	False alarms are counted and published	Reported every season

1.6 Sustainable Development Goals

[**Project Name**] contributes to three SDG targets [22]: **11.5**, reduce deaths and people affected by disasters, including water-related disasters, the project's main goal; **13.1**, strengthen resilience to climate-related hazards such as the intense rainfall of 2018 and 2026; and **1.5**, reduce the exposure of the poor and vulnerable to extreme events, through volunteers and the assistance flag (Section 1.7). It also supports *Early Warnings for All* [11] and Action 8a of the national strategy [8].

1.7 Inclusion and local actors

Warnings that reach everyone. The same warning travels through several channels, so it reaches people whatever their age, ability or access to a phone: a flashing beacon (for people who cannot hear) and a siren (for people who cannot see) at the crossing, needing no network or phone; SMS in Arabic or French with a place and an action, readable on any basic phone; and volunteers, alerted by SMS or a radio handheld that works without the mobile network, who pass the instruction on in person to elderly residents and people without a phone. Volunteers from community groups often raise risk awareness better than officials [23]. A household can also ask, through an optional assistance flag, to be visited first. Joining is voluntary: the registry holds only a phone number, a

location, a language and the flag, with written consent, stored locally under Tunisia's data-protection law [24].

Local actors. The *Nabeul municipality* hosts the gateway, names the duty officer, and trains technicians to inspect nodes and swap parts with a written manual. *Civil Protection (ONPC)*, present through 24 regional directorates and 196 brigades [1], receives escalated alerts and validates message wording. The *regional committee under the governor* decides on public alerts under Law 91-39 [1]; it receives our evidence and keeps the final decision. *INM* and the *Oued Souhil experimental station* are possible data partners [10, 14], and volunteers and residents help choose node locations, test messages and take part in drills. Parts are locally replaceable, the software is open source and SMS goes only to a small registered group, so running costs are mostly maintenance time.

1.8 Requirements

The evidence above sets six requirements (Table 1.3); Chapter 2 traces every design choice and test back to them.

Table 1.3: Requirements and the evidence behind them.

ID	Requirement	Why (evidence)
R1	Work without mains power, internet or mobile network	Storms that cause flash floods also cut power and networks; bulk SMS can be delayed by congestion [7].
R2	Confirm receipt by a responsible person, and act when nobody answers	National SMS and cell broadcast have no back channel [19, 7]; alert dissemination is a named weakness [1].
R3	Give a safe action, not just a danger level	In 2018 a warning existed the day before, yet some residents stayed on rooftops [3].
R4	Limit false alarms	False alarms reduce trust in the next warning [25].
R5	Be low-cost and locally repairable	The national strategy calls for simple, low-cost equipment [8].
R6	Fit the institutional chain; never replace human decisions on public alerts	Public alerts are decided by the national and regional committees [1, 19].

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